

Roll No. | Total No. of Pages : 4

ME-1850

B. Tech. (First Semester)

EXAMINATION, 2025-26

MANUFACTURING PROCESS

Time : Two Hours

Maximum Marks : 30

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *five* questions. Each question carries 2 marks. $5 \times 2 = 10$

1. Explain Malleability.
2. Explain the application of Brazing.

P. T. O.

3. What is shrinkage allowance ?
4. What is Elasticity ?
5. Explain strength and toughness.
6. Give the names of various moulding sands.
7. Explain any *one* casting defect.
8. Discuss Pig Iron and Wrought Iron.
9. Write composition and properties of gray cast iron.
10. Discuss the properties and composition of white cast iron.

Section—B

(Short Answer Type Questions)

Note : Attempt any *two* questions. Each question carries 10 marks. $2 \times 5 = 10$

1. Explain heat treatment process. Define stages of heat treatment. Describe annealing and normalizing.

2. What is Metal Forming ? Explain rolling with its type and discuss wire drawing operation.
3. Differentiate between Hot Working Properties and Cold Working Properties.
4. What is Casting ? What is Moulding Sand ?
Also discuss the types of moulding sand.

Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question in maximum
300 words. Each question carries
20 marks.

$1 \times 10 = 10$

1. Explain the principle of lathe machine and give name of parts of lathe machine with a block diagram. Also discuss any *five* lathe operations.

2. What is Welding ? Explain oxy-acetylene gas welding with a block diagram.
3. Write comparison between AC and DC Welding.
4. What is Metal Forming ? Explain any *two* metal forming processes with their application in detail.

× × × × ×

Section—C

(Long Answer Type Questions)

Note : Attempt any one question in maximum

300 words. Each question carries

10 marks.

30 marks.

Explain the principle of lathe machine and

give name of parts of lathe machine.

block diagram. Also draw a

ME-1850(O)

B. Tech. (First Semester) EXAMINATION, 2024-25 MANUFACTURING PROCESS

Time : Two Hours

Maximum Marks : 30

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *five* questions. Answer each question in maximum **30** words.
Each question carries 2 marks. $5 \times 2 = 10$

-1. Explain ductility.

P. T. O.

2. What are ferrous metals and non-ferrous metals ?
- 3. What is Malleability ?
4. What is Toughness ?
- 5. What is Elasticity ?
6. Give the properties and composition of bronze.
7. What is stiffness ?
8. What is smelting ?
9. Write the properties of zinc.
- 10. What is creep ?

Section—B

(Short Answer Type Questions)

Note : Attempt any *two* questions. Each question carries 5 marks. $2 \times 5 = 10$

1. What is extrusion ? Explain its types with diagram.
- 2. Explain difference between shaper and planner.
- 3. What is casting ? Explain the types of moulding sand.

[3]

4. What is soldering and brazing ?

Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question. Each question carries 10 marks. $1 \times 10 = 10$

1. What is arc welding ? Differentiate AC and DC welding.
- 2. Describe the working principle of lathe machine with block diagram.
3. What is heat treatment ? Explain all types of heat treatment processes.
4. Explain stress strain diagram for ductile material.

Roll No. ~~.....~~ [Total No. of Pages : 3

ME-1850

B. Tech. (First Semester)

EXAMINATION, 2020

MANUFACTURING PROCESS

Time : Two Hours

Maximum Marks : 30

Note : Attempt questions from both Sections as directed.

Section—A

(Short Answer Type Questions)

Note : Attempt any six questions. Each question carries 2 marks.

$6 \times 2 = 12$

1. Define the following :

(a) Brittleness

2. Explain any *one* method of making metal powder.
3. What are the core material properties required for a making a good core ?
4. What is the difference between shaper and planar machine ?
5. What is Extrusion process ? What are its types ?
6. What is the difference between hot working and cold working ?
7. What is steel ? Explain its different types in short.
8. Why is heat treatment of steel required ? Explain any *one* type of heat treatment process.
9. What is Brass ? Discuss its applications.

Section—B

(Long Answer Type Questions)

Note : Attempt any *two* questions. Each question carries 9 marks.

$$2 \times 9 = 18$$

- 1/ Discuss Oxyacetylene Gas welding in detail.

2. Explain different types of forging processes in detail with a neat figure.
3. Why are the basic requirements for choosing a plant location ? Explain different types of plant layout.
4. Explain different zones of Cupola furnace with neat figure.

ME-102/1850

B.Tech. (Semester-I) Examination-2017
Manufacturing Process

Time: Three Hours

Maximum Marks: 50

Note: Attempt questions from all the sections.

Section-A

(Short Answer Type Questions)

Note: Attempt any five questions. Each question carries
4 marks. (4x5=20)

1. ~~Briefly explain what do you understand by toughness?~~

2. Describe puddling process.

3. Briefly explain rolling process.

4. What is manufacturing?

5. Describe welding and different type of welding process.

6. Define malleability and ductility.
7. Write short notes on spot welding.
- ~~8.~~ Write notes on plant layout.

Section-B

(Long Answer Type Questions)

Note: Attempt any three questions. Each question carries 10 marks. (10x3=30)

1. Write notes on production and productivity.
- / 2. Give main objectives and advantages of good plant layout.
- / 3. Give principle of oxy-acetylene welding with neat diagram.
4. Describe cupola furnace with neat sketch diagram.
- ↓ 5. What is grinding? Give specification of wheel.